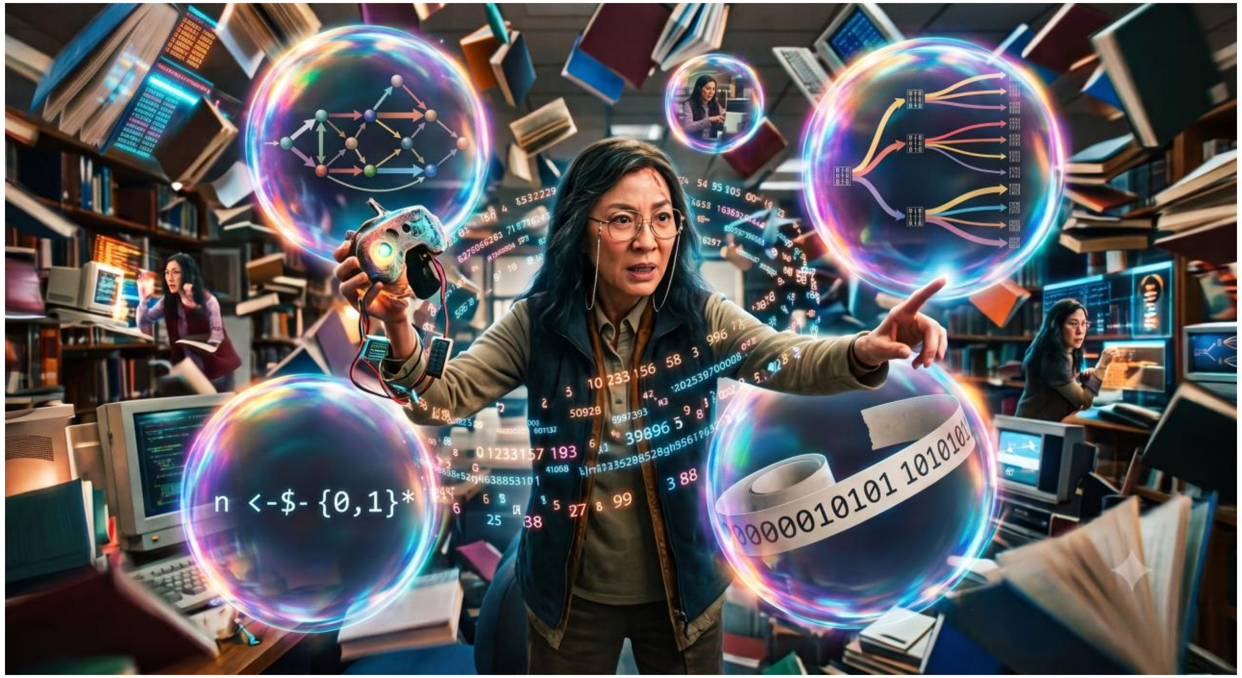
Everything Everywhere All at “Nonce” :
확률적 멀티버스 속 암호 프로토콜 검증



손병호, 배경민 (포항공대)

제 11회 소프트웨어재난연구센터 워크샵 (대구)

2026년 7월 14일

암호 프로토콜 검증은 매우 중요하고 **어렵다**

암호 프로토콜 검증은 매우 중요하고 어렵다

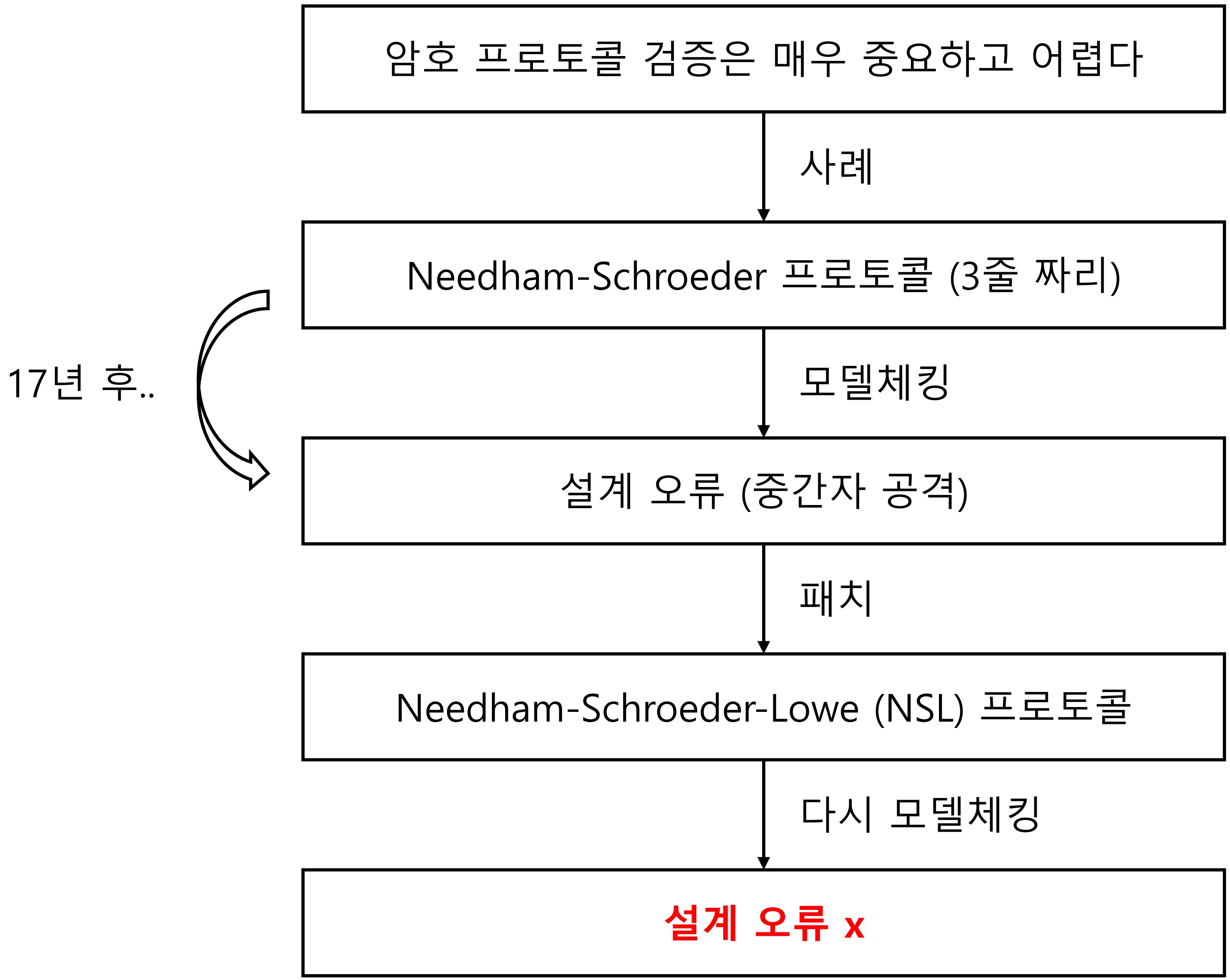
↓ 사례

Needham-Schroeder 프로토콜 (**3줄 짜리**)

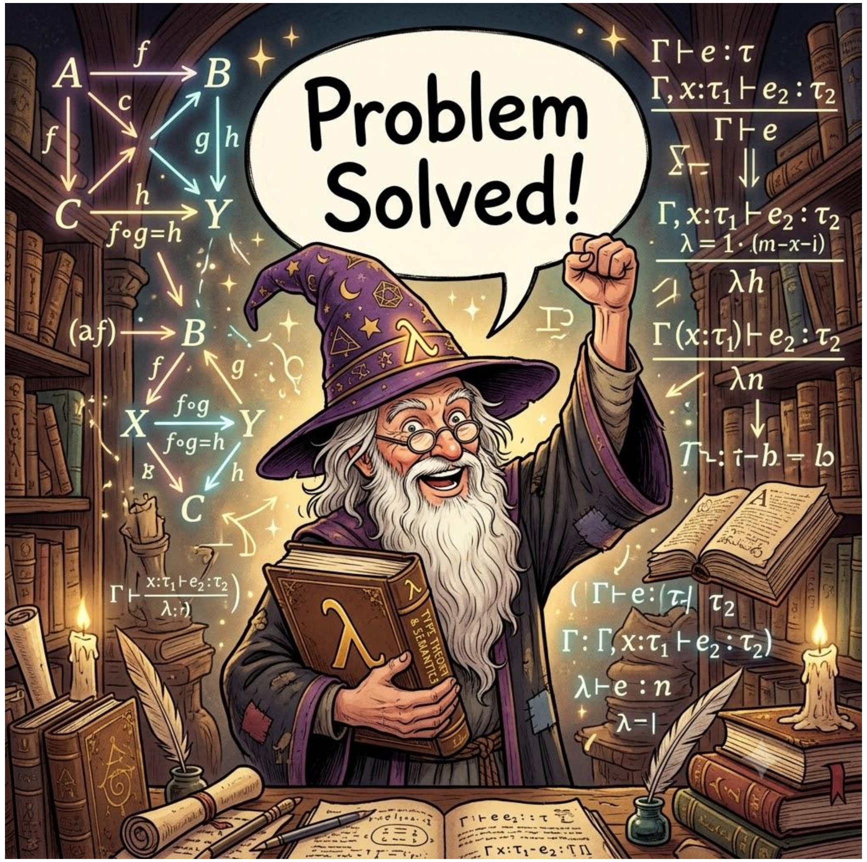
↓ 모델체킹

17년 후..

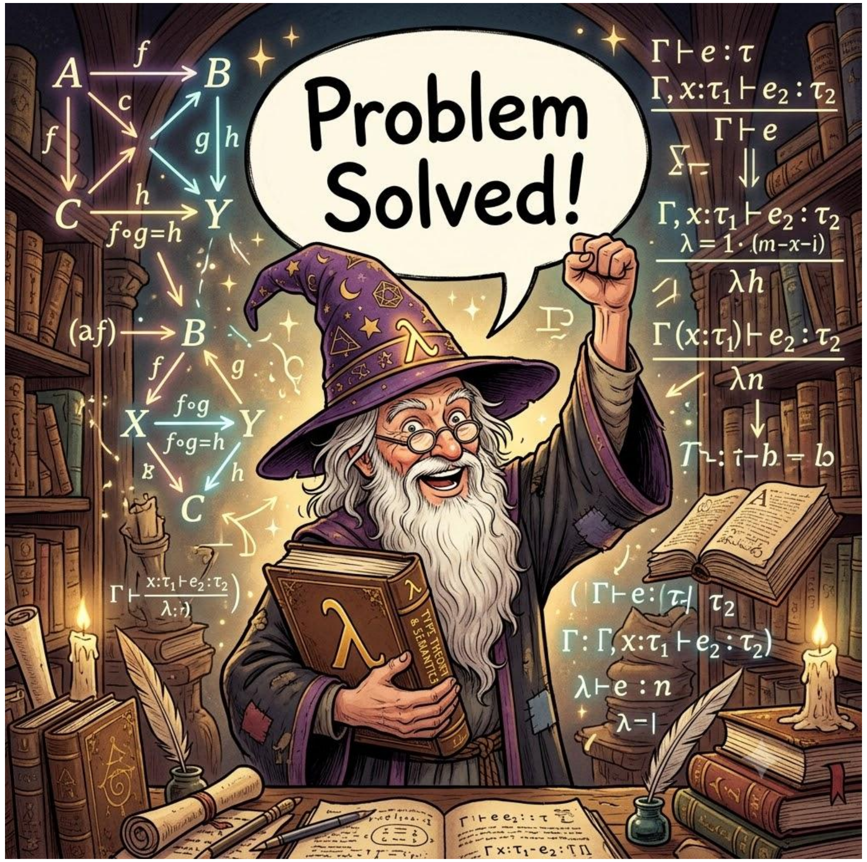
설계 오류 (중간자 공격)


암호 프로토콜 검증은 매우 중요하고 어렵다
사례
Needham-Schroeder 프로토콜 (3줄 짜리)
모델체킹
17년 후..
설계 오류 (중간자 공격)
패치
Needham-Schroeder-Lowe (NSL) 프로토콜
다시 모델체킹
설계 오류 x

논리학자 :


Problem Solved!
$\Gamma \vdash e : \tau$
$\Gamma, x{:}\tau_1 \vdash e_2 : \tau_2$
$\Gamma \vdash e$
$f \circ g = h$

암호학자 :



암호학자의 관점

(비트의 세계)

메시지는 **유한한 비트스트링**이다

(확률의 세계)

공격자는 **난수**를 생성할 수 있다

⇩

공격자는 운 좋게 비밀 메시지를 때려 맞출 수 있다!

공격자에게 무한한 시간이 있다면?

공격 성공확률 : 0.000000000000000000000000000001

무한한 시간
(증폭)

공격 성공확률 : 0.9999999999999999999999999999999

현실적 공격자

공격 성공확률 : 0.00000000000000000000000000000001

다항시간

공격 성공확률: **???**

암호학적 증명

확률 이론　　계산복잡도 이론

다항시간 공격자의 “**성공확률**”이 “**얼마나 빨리**” 1에 수렴하는가?

PROOF Let Π denote the El Gamal encryption scheme. We prove that Π has indistinguishable encryptions in the presence of an eavesdropper; by Theorem 10.10 this implies that it is CPA-secure.

Let $\mathcal{A}$ be a probabilistic, polynomial-time adversary, and define

$$\varepsilon(n) \stackrel{\text{def}}{=} \Pr[\mathsf{PubK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(n) = 1].$$

Consider the modified “encryption scheme” $\widetilde{\Pi}$ where Gen is the same as in Π, but encryption of a message m with respect to the public key $\langle \mathbb{G}, q, g, h \rangle$ is done by choosing random $y \leftarrow \mathbb{Z}_q$ and $z \leftarrow \mathbb{Z}_q$ and outputting the ciphertext

$$\langle g^y,\ g^z \cdot m \rangle.$$

Although $\widetilde{\Pi}$ is not actually an encryption scheme (as there is no way for the receiver to decrypt), the experiment $\mathsf{PubK}^{\mathsf{eav}}_{\mathcal{A},\widetilde{\Pi}}(n)$ is still well-defined since the experiment depends only on the algorithms for key generation and for encryption.

Lemma 10.18, and the discussion that immediately follows it, imply that the second component of the ciphertext in scheme $\widetilde{\Pi}$ is a uniformly-distributed group element and, in particular, is independent of the message m being encrypted. The first component of the ciphertext is trivially independent of m. Taken together, this means that the distribution of the ciphertext is independent of m and hence contains no information about m. It follows that

$$\Pr[\mathsf{PubK}^{\mathsf{eav}}_{\mathcal{A},\widetilde{\Pi}}(n) = 1] = \frac{1}{2}.$$

Now consider the following PPT algorithm D that attempts to solve the DDH problem relative to $\mathcal{G}$:

Algorithm D:
The algorithm is given $\mathbb{G}, q, g, g_1, g_2, g_3$ as input.

- Set $pk = \langle \mathbb{G}, q, g, g_1 \rangle$ and run $\mathcal{A}(pk)$ to obtain two messages m_0, m_1.
- Choose a random bit b, and set $c_1 := g_2$ and $c_2 := g_3 \cdot m_b$.
- Give the ciphertext $\langle c_1, c_2 \rangle$ to $\mathcal{A}$ and obtain an output bit b'. If $b' = b$, output 1; otherwise, output 0.

Let us analyze the behavior of D. There are two cases to consider:

Case 1: Say the input to D is generated by running $\mathsf{Gen}(1^n)$ to ob $(\mathbb{G}, q, g)$, then choosing random $x, y, z \in \mathbb{Z}_q$, and finally setting $g_1 :=$ $g_2 := g^y$, and $g_3 := g^z$. Then D runs $\mathcal{A}$ on a public key constructed as

$$pk = \langle \mathbb{G}, q, g, g^x \rangle$$

and a ciphertext constructed as

$$\langle c_1, c_2 \rangle = \langle g^y,\ g^z \cdot m_b \rangle.$$

We see that in this case the view of $\mathcal{A}$ when run as a subroutine by distributed exactly as $\mathcal{A}$'s view in experiment $\mathsf{PubK}^{\mathsf{eav}}_{\mathcal{A},\widetilde{\Pi}}(n)$. Since D outpu exactly when the output b' of $\mathcal{A}$ is equal to b, we have that

$$\Pr[D(\mathbb{G}, q, g, g^x, g^y, g^z) = 1] = \Pr[\mathsf{PubK}^{\mathsf{eav}}_{\mathcal{A},\widetilde{\Pi}}(n) = 1] = \frac{1}{2}.$$

Case 2: Say the input to D is generated by running $\mathsf{Gen}(1^n)$ to obtain $(\mathbb{G}, q, g)$, then choosing random $x, y \in \mathbb{Z}_q$, and finally setting $g_1 := g^x$, $g_2 := g^y$, and $g_3 := g^{xy}$. Then D runs $\mathcal{A}$ on a public key constructed as

$$pk = \langle \mathbb{G}, q, g, g^x \rangle$$

and a ciphertext constructed as

$$\langle c_1, c_2 \rangle = \langle g^y,\ g^{xy} \cdot m_b \rangle = \langle g^y,\ (g^x)^y \cdot m_b \rangle.$$

We see that in this case the view of $\mathcal{A}$ when run as a subroutine by D is distributed exactly as $\mathcal{A}$'s view in experiment $\mathsf{PubK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(n)$. Since D outputs 1 exactly when the output b' of $\mathcal{A}$ is equal to b, we have that

$$\Pr[D(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] = \Pr[\mathsf{PubK}^{\mathsf{eav}}_{\mathcal{A},\Pi}(n) = 1] = \varepsilon(n).$$

Since the DDH problem is hard relative to $\mathcal{G}$, there must exist a negligible function negl such that

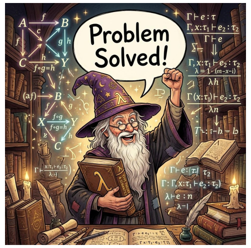
$$\mathsf{negl}(n) = \left| \Pr[D(\mathbb{G}, q, g, g^x, g^y, g^z) = 1] - \Pr[D(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] \right|$$
$$= \left| \frac{1}{2} - \varepsilon(n) \right|.$$

But this implies that $\varepsilon(n) \leq \frac{1}{2} + \mathsf{negl}(n)$, completing the proof. ■

John Katz, Introduction to Modern Cryptography

두 도시 이야기

검증
용이성

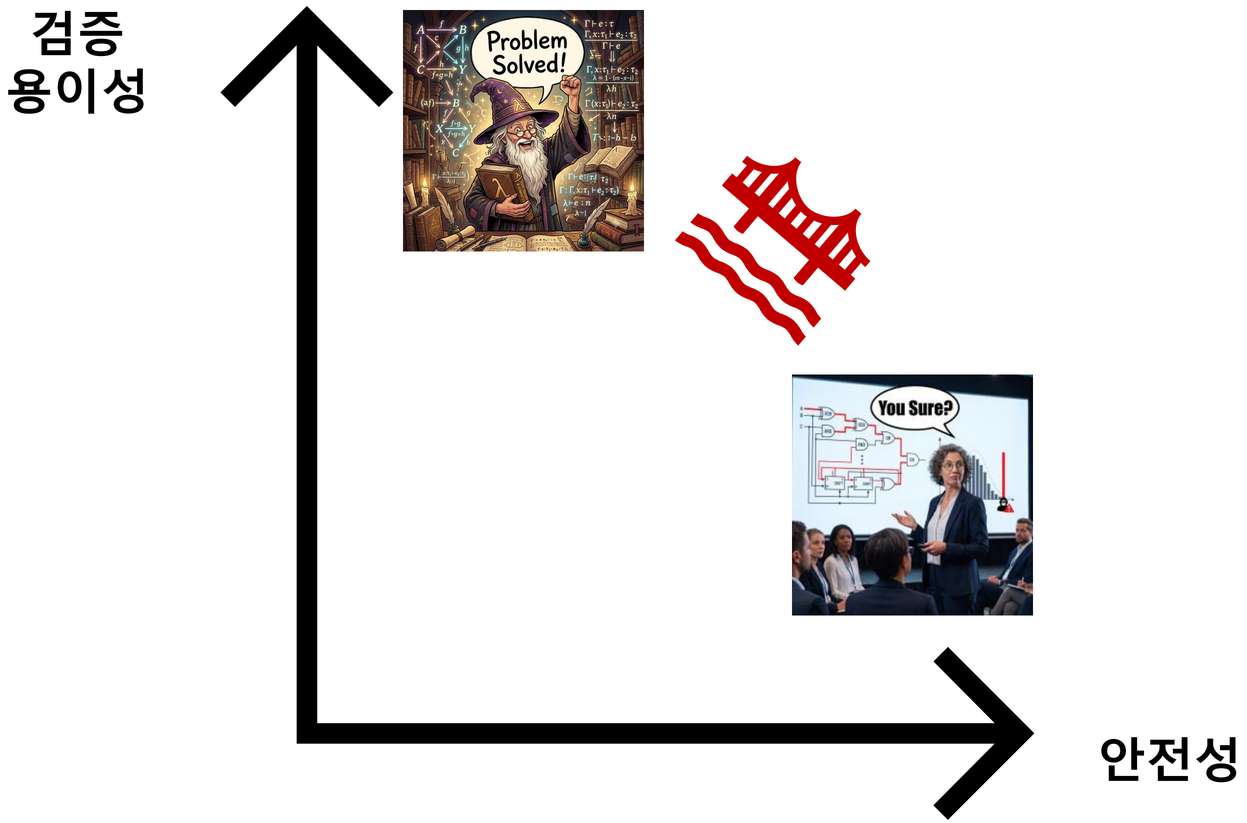


(모델체킹)

(암호학)

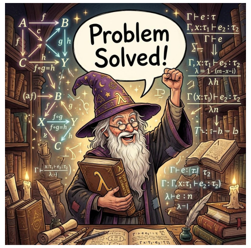
안전성

연구 목표 : 두 도시 잇기



기존 기술

검증
용이성



Université
de Rennes

UMR IRISA

Squirrel Prover

안전성

Squirrel의 표현력

Squirrel Prover

(∀) “모든 도달가능한 실행경로에서 논리식 ϕ가 대부분의 확률로 참” **(확률 공격의 부재)**

$\not\Leftrightarrow$

(∃) “어떤 도달가능한 실행경로에서 논리식 ϕ가 유의미한 확률로 거짓” **(확률 공격의 존재)**

더 높은 표현력

LEAN THEOREM PROVER

Squirrel Prover

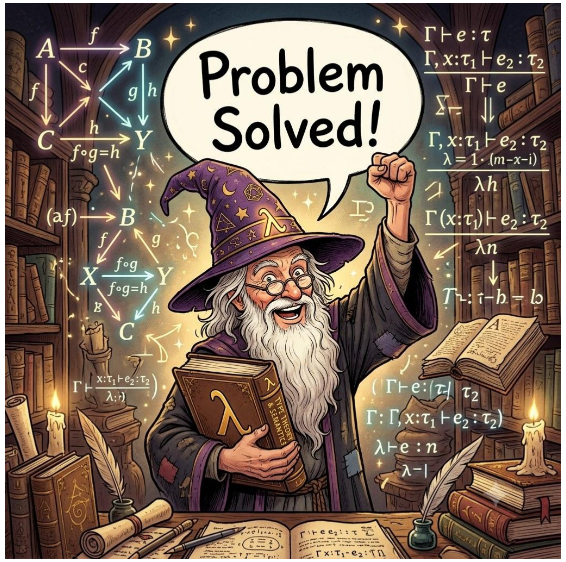
(∀) “모든 도달가능한 실행경로에서 논리식 ϕ가 대부분의 확률로 참” **(확률 공격의 부재)**

$\not\Leftrightarrow$

(∃) “어떤 도달가능한 실행경로에서 논리식 ϕ가 유의미한 확률로 거짓” **(확률 공격의 존재)**

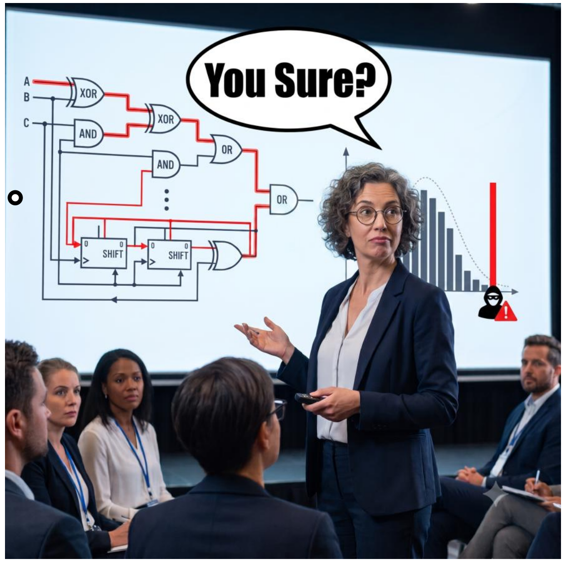
문제 : 명세할 게 너무 많음

“어떤 도달가능한 실행경로에서 논리식 ϕ가 유의미한 확률로 거짓” (공격의 존재)



참 / 거짓

확률이
blah blah

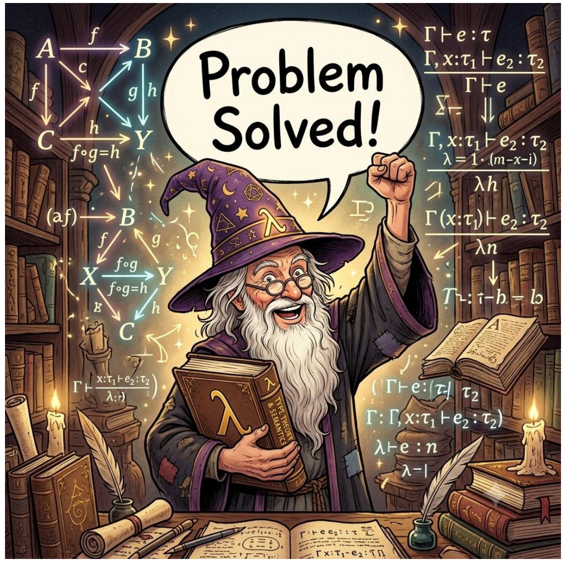


동시성 모델 일차논리

튜링머신 계산복잡도 확률론

LEAN THEOREM PROVER

만병통치약 : 요약



참 거짓

유의미한 확률로 참

대부분 확률로 참

동시성 모델 일차논리

암호학

튜링머신 계산복잡도 확률론

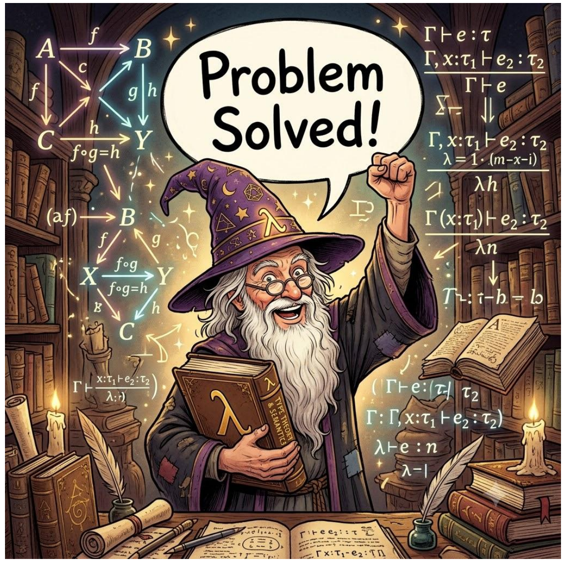
양상논리 : (확률적) 멀티버스 속 참 개념

"이 확률세계에서 ◇□p가 참이다" ⇔ "p가 유의미한 확률로 참이다"

"이 확률세계에서 □p가 참이다" ⇔ "모든 근처 확률세계에서 p가 참이다"

"이 확률세계에서 ◇p가 참이다" ⇔ "어떤 근처 확률세계에서 p가 참이다"

줄어든 명세 범위



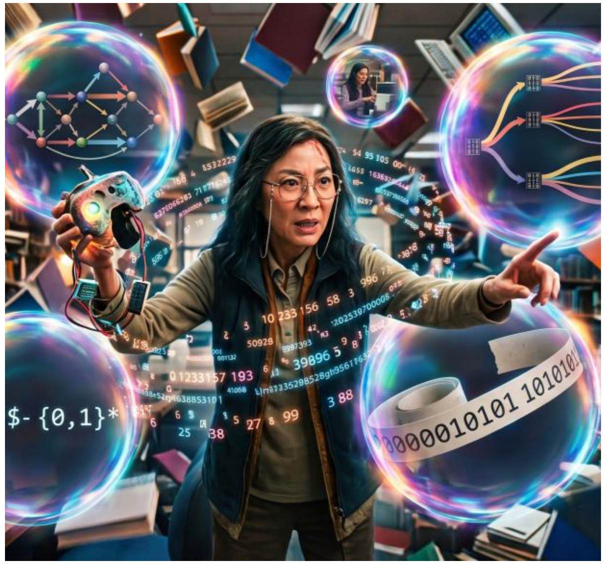
$\Box(A \to B) \to (\Box A \to \Box B)$

$\Box A \to \Diamond A$

$\Box A \to A$

$A \to \Box\Diamond A$

$\Diamond A \to \Box\Diamond A$



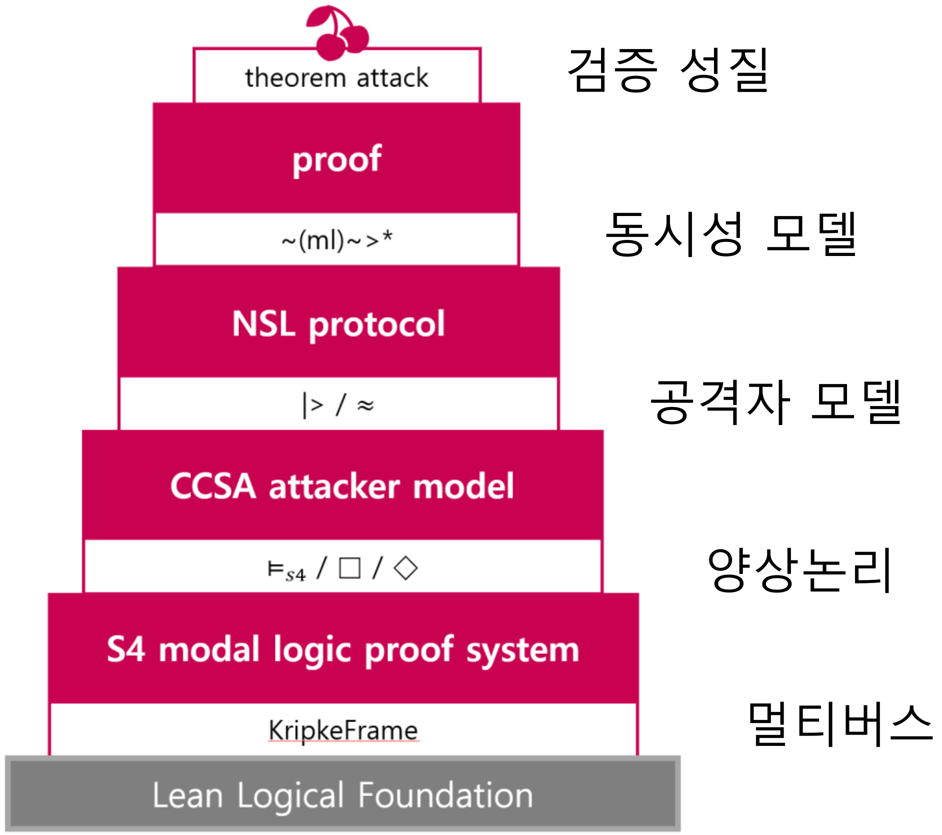
동시성 모델 일차논리

양상논리

확률적 멀티버스

LEAN THEOREM PROVER (고차논리)

5단 요약 케이크 in Lean



결과 : 확률 없이 확률공격 증명 성공

“어떤 도달가능한 실행경로에서 논리식 ϕ가 유의미한 확률로 거짓” (공격의 존재)

⇕

```
theorem attack : ∃ st ml, (conf0 ~(ml)~>* st) ∧ ∃ (w : K.World), w ⊨s4 (st.cond ⋏ □◇(st.chan |> nB 0))
```

“실행경로” “멀티버스” “유의미한 확률로 참”

향후 계획

- 더 높은 자동화
  - 모델체킹 알고리즘을 Tactic으로 지원

감사합니다!